M. SHAHID MODI

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EDUCATION

Rensselaer Polytechnic Institute, Troy, NY *Since Jan '21*
Ph.D. Computer Science; *Advisor: Dr. Boleslaw Szymanski; GPA 3.5*
Thesis work: Agentic Interaction AI for human-subject research, AI for large-scale analyses (training, finetuning, alignment), Social Media Community Modeling and Simulation
Project-based expertise: Finetuning LLMs with parameter efficient techniques, Human-AI interaction, Computer Vision, Natural Language Processing, Large dataset curation, Experiment design for human-subject research
Obtained M.S. CS @ RPI *('21-'23); Ph.D. graduation expected Aug 2026*

Dr. M.G.R. Edu & Research Institute, India *2016 - 2020*
B.Tech., Computer Science & Engineering; CGPA 9.29/ 10 (Hons)

PUBLICATIONS

1. Modi, M.S. & Szymanski, B.K. ***An LLM-Encoder Ensemble for Coarse-to-Fine Valence-Arousal Sentiment Prediction.*** *ACL SemEval* (Under review) (2026)
2. Szymanski, B.K., Zhang, Y., Uzzi, B. & Modi, M.S. ***Analysis of Collaboration in CS Prizewinning with a Nobel-Turing Comparison***. *arXiv.2512.23919* (Submission stage) (2026)
3. Flamino, J., Modi, M.S., Szymanski, B.K., Cross, B. & Mikolajczyk, C. **Testing the *Limits of Large Language Models in Debating Humans.*** *Nat Sci Rep,* 15, 13852 (2025).
4. Hu, J.C., Modi, M.S. & Szymanski, B.K. ***Using LLMs to Infer Non-Binary COVID-19 Sentiments of Chinese Microbloggers.*** *Entropy*, 27(3), 290 (2025).
5. Modi, M.S., Flamino, J. & Szymanski, B.K. ***Dynamics of Ideological Biases of Social Media Users***. *IEEE Communications Magazine,* 62, 36–42 (2024).

RESEARCH

LLMs as Active Agents in Conversational Contexts (Ongoing long-term project) *Since Jun '23*

- Conceptualized and co-led **human subject research study** from proposal stage to study the behavior of human debaters against LLMs impersonating humans.
- Built an LLM-integrated messenger app and automated scheduling for online debates with **over 100 participants** and **over 100 LLM bots.**
- Currently building **human-like LLM agents** with **emotion mirroring** through **dataset curation**, **supervised finetuning** (SFT), and knowledge graphs for **RAG.**
- Achieving more aligned agents through **parameter-efficient finetuning** on **quantized models** and **Direct Preference Optimization** (DPO).

Longitudinal Analysis of Social Media Network Dynamics (Ongoing) *Since Jun '21*

- Processed **large social media datasets (60GB+)** and performed comparative analysis of community formation through political biases.
- Found a news echo chamber on the Parler social network using **community detection algorithms** and **natural language processing techniques** for first paper.
- Designed processing, analysis and visualization pipelines in Python and Gephi. Project won best masters poster presentation in Fall 2022 at RPI's grad poster event.

Object Detection using YOLOv5 on Aerial Elephant Dataset (AED) *Fall '23*

- Led a group of four students on a computer vision project drawing bounding boxes around elephants who blend in with their environment, appearing as tiny dots.
- Identified 56% of elephants on limited hardware with a **YOLOv5 ML model** using computer vision techniques to account for environmental factors such as ground-sampling distance and background.

Anti-Adversarial GNN – Membership Inference Attack (MIA) Defense *Fall '22*

- Developed a Model Membership Inference Attack defense technique called **NodeDrop** using **Graph Neural Networks** for research-based coursework.
- Used technique to fill gap in Graph Neural Net's weaknesses in the learning stage, **preventing training data being leaked to attackers** by using a regularization step.
- Presented this novel application of regularization to MIA defenses for GNNs as final project for RPI's Trustworthy ML course.

TECHNICAL SKILLS

- **Machine Learning** – Supervised finetuning of LLMs, Preference Optimization (DPO, PPO), Retrieval Augmented Gen (RAG)**,** Synthetic data generation for fitting and scaling LLMs, Training ML models (Transformers, YOLO, Faster RCNN, Graph NN)
- **Trustworthy ML & Computer Vision** – Adversarial ML and defense techniques, regularization and optimizations**,** image data preprocessing, object detection, recognition and tracking in image datasets
- **Computational Social Science** – LLM-Human communication, community dynamics, statistics, inference and predictive modelling, conversational dynamics
- **Python** – Pandas, OpenCV, PyTorch, TensorFlow, NumPy/SciPy, Visualizations
- **C++ parallel programming** – Projects on RPI's AiMOS supercomputer
- **Software Dev** – Node.js, PostgreSQL, SQLite, Flet, Flutter, Git version control, Agile development

WORK EXPERIENCE

Research Assistant, Network Science and Technology Center *Since Jun '21*

- Working in my advisor's lab on research funded by NSF and DARPA grants and responsible for meeting publication expectations and maintaining timelines.
- Mentored and delegated tasks to 3 undergraduate researchers, collaborating with them to advance research goals on multiple projects.
- Brainstorming and writing new proposals and white papers for grant applications, including for NSF, DARPA, DoD and Army Research Lab grants.

Teaching Assistant, Rensselaer Polytechnic Institute *Since Jan '21*

- Assisted four different classes at RPI over three years which included delivering some lectures, leading recitations, holding office hours and meetings, and grading.
- Courses: Introduction to Algorithms, Application Design using Java, Software Design and Documentation (several times), Open Source Software

Fullstack Developer Intern - Wizr Wealth *May '20 - Dec '20*

- Developed a Fullstack webapp aimed at connecting investors with finance experts using Flutter (Android/ iOS cross-platform) frontend and JS/SQL backend.
- Worked with a UI/UX expert and delivered rapid prototypes until completion.